Introductory activity

Activity 1 · A network of bush taxis 40 min . Five towns A, B, C, D, E are linked by direct bush-taxi routes, in both directions: A-B, A-C, A-D, B-C, C-D and D-E.

1. Represent the network by a diagram: one point per town, one line per route.
2. For each town, count the number of routes leaving it. Compute the sum of these numbers and compare it with the number of routes. Explain.
3. List the journeys from A to C that use exactly two routes (a town may be visited again). How many are there?
4. Same question for the journeys from A to E using exactly three routes.
5. We store the information in a table M with 5 rows and 5 columns: the entry in row X and column Y is 1 if there is a direct route between X and Y, 0 otherwise. Write M, then compute M^2 and M^3 with a calculator. Compare some entries with questions 3 and 4.
6. State a conjecture.

Observing is not proving: the properties conjectured here are established in the course.